A Novel Fault Self-Detectable Universal Quantum Reversible Circuits Array Design

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***Abstract*—Many classic logic gates are irreversible, such as "And gate", or "Or gate". This irreversibility of the calculation results in energy consumption. Reversible circuit technology, like quantum circuits, in which the entire calculation process does not consume energy, can be used for low-power microchip and quantum communication applications. Most technologies focus on the synthesis of reversible circuits, rarely on fault testing. Fault testing is certainly an important step in any powerful implementation. In this paper, we design two reversible quantum units that each one can be used as testing circuit for another. The proposed design for fault self-detection is based on two facts. The complex quantum circuit can be achieved by combining the universal quantum units in parallel or cascade. The quantum circuit is naturally reversible. As a result, a novel universal quantum unit pair for fault self-detection is designed. The proposed structure does not need any extra DFT testing circuit; therefore, the testing cost has been reduced. Besides, even the scale of the circuit increases, the testing complexity stays in O(√*n*). Thus, our proposed design is efficient for a large scale quantum circuit.**

Keywords— Quantum logic array; Reversible Circuit; Low Power circuit, Single Stuck-at-Faults; Fault detection

I. INTRODUCTION

Quantum computing is a state of art computation that is completely different from the classical computation. Its most important superiority is reflected in the quantum parallel computing. Quantum computation is implement by quantum computers, which through the quantum logic gates to control and manipulate the evolution and transfer of quantum states for quantum information processing [1]. Quantum logic gates process the information logically [2]. One attractive feature in quantum computing promises the reversibility, which means the quantum circuits are reversible. It is established by Landauer's principle [3], "erasing information, as opposed to writing information, is what causes energy loss and heat" in computational process." The common CMOS VLSI circuits are not reversible; as a simple NAND gate for example, it has two input bits of information and one output bit of information. From two input bits to one output bit, one bit of information is erased (loss of information) and the computing process produces heat. Due to this "heating issue", the traditional circuits has the limitation to build high-dimensional system. Therefore, reversible circuits are the basis of not only quantum computing science, but also low-power CMOS and nanotechnology research. Quantum logic gates meet the requirements of reversible computing and have the characteristics of reversible operation. There is no bit loss of information from conversion between electric energy and thermal energy, so it is essentially solving the problem of computer energy consumption[4].

Furthermore, it is necessary to ensure the function of the quantum logic circuit is correct; thus guarantees the function of the quantum computer can be correctly implemented. Therefore, it is important to test, detect and locate the fault in quantum circuits on time since the circuits cannot be always completely reliable in lifetime. There are many reasons why a quantum logic circuit may fail. Like traditional circuits, the logic faults of a quantum circuit can be classified as a stuck at fault model, a broken control model, Missing-gate fault model, redundant-gate faults model, adding an appearance fault model, *etc*.[5] . Currently, there is no single method to detect all types of faults in a logic circuit even there are some existing testing methods [6], such as Integer linear programming algorithm (ILP)[7], Design for test (DFT) algorithm [8], Automatic test pattern generation (ATPG) algorithm [9]. ATPG, for example, as its name suggests, is an electronic design tool for generating input patterns that can verify whether the circuit has faults at some/certain locations. One common method of finding the input patterns is to randomly generate the inputs. It is efficient in a small circuit, but not efficient when the scale goes large. For these very large scale integrated circuit test platforms, due to the high complexity of the circuits, it is extremely difficult to achieve a very high faults coverage. That is, different ATPG algorithm must be designed for testing different types of digital circuits. Thus, one testing algorithm is generally defined for a specific type of fault detection and usually needs a DFT circuit either connected outside or embedded in the chip for fault testing. Accordingly, it is not possible to solve all detection problems.

Is it possible to detect the fault by using the logic circuit itself; therefore, we can eliminate the DFT circuits for testing? The answer could become clear when it goes to a quantum circuit. It is easy to design a universal quantum logic gate and further use them to build much more complicated functional logic circuits [10-12]. These universal quantum logic gate can be tested by adding its reverse gate (*e.g* quantum Tofolli gate is an example of universal logic gate and is self-reversible gate). In this paper, we introduce a novel fault self-detectable universal quantum unit array. We used single stuck-at-fault model for simulation in the paper. The proposed quantum unit array does not need extra DFT circuit for testing and has an efficient testing for a large scale of quantum circuit system.

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II. PROBLEM IDENTIFICATION

Quantum logic circuit is one of the key issues in quantum computer and quantum information technology research. Firstly, quantum logic gate could be functionally universally. One example is the classical quantum gate named Tofolli gate [13]. Feeding different values to controlling-bits, it eventually allows a NAND, XOR, and INV gate[14]. They are basic blocks for building more complicated functional circuits. Figure 1 shows the examples of quantum Tofolli gates with their corresponding traditional logic gates. Reversible logic is a logic that implements the *n*-input and *n*-output function and each input maps to a unique output. The number of input lines in the reversible circuit is the same as the number of output lines [15]. Based on the properties of reversible logic, we can determine that there is no feedback, fan-in, and fan-out in the reversible circuit, so the reversible network consists of a cascade of each reversible circuit [16].

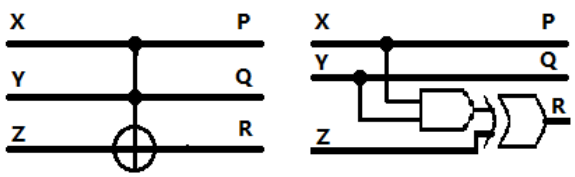


Figure 1. Quantum Tofolli gates
($P = X$; $Q = Y$; $R = (Y \bullet X) \oplus Z$)

There are many possible faults in a quantum circuit. The one-point error models are well known such as stuck-at fault model, missing-gate fault model, and broken control model. As a stuck-at fault in a common circuit, the value at one or more connection lines is always a logic 1 or 0. By contrast, it appears that the value at one line is stuck at 1 or 0 in a quantum circuit. There are two types of stuck-at-fault models: (1) a single stuck at fault model (SSFM); and (2) a multiple stuck-at-fault model (MSFM). In a single stuck-at-fault model, a signal line in a circuit is stuck at a certain logic value. If the circuit has n signal lines, there are 2×n possible single stuck-at-faults. The stuck failure model is a logical failure model whose impact is permanent. In this paper we mainly focus on the single stuck-at-fault model problem because the single stuck-at fault (S@F) is the most common type and many other type of fault models are usually can be treated as the single stuck-at fault model problem [17, 18] .

As the common method to detect the fault in traditional digital circuit, the complete test set usually needs to be defined at the beginning. The complete test set includes all possible input patterns that can cover all faults in the logic circuit. In addition, if the subset with the minimal number of patterns in a complete test set can detect all faults in the logic circuit, it is known as the minimal complete test set [19]. Second step is feeding the minimal complete test set as the inputs. Thus, the circuit under test is a type of black box testing. If one output is different from the expected value by comparing the truth table and the fault table, we know there is a fault in the circuit. For an n-input digital circuit, if there are *m* lines, the possible single-stuck faults are *2m* and all testing patterns are n^2. Thus, if the number of circuit lines and number of inputs increases gradually, the testing complexity increases exponentially. Obviously, this is not an efficient method to test fault for a large scale logic circuit without a proposed DFT circuit. In the paper, the designed quantum unit is a universal block for both performing function and testing, thus we do not increase the lines and the inputs. However, if the number of units in the circuit keeps increasing, the testing complexity will increase linear. This is still too harming to do the testing in a large scale circuit.

To solve the above problems, the first purpose of the design in the paper is the quantum reversible circuit – it is a low power design. Second, the designed universal quantum unit pair that includes unit A and unit B. Unit A and B are reversed circuits to each other; moreover, both of them are universal circuits for building more complex functional logic in the array. Finally, the proposed universal quantum unit array does not require extra DFT circuit for fault detection. In addition, it has a self-detection mode to not only find the fault but also locate the unit where the fault occurs in an efficient way.

III. RELATED WORK

Circuit fault detection is the detection of unwanted events. We can quickly test the logic circuit by inputting the test vector set to determine if it is faulty. In recent years, the detection of quantum reversible circuits has been emphasized. Some researchers have devoted themselves to the research of fault detection of reversible circuits. There are three main detection methods [20]: 1). Offline test; 2). Online test; 3). Fault tolerance. Offline test refers to verifying the output data after the circuit under test implements its function. If the output data is not the same, it indicates that the logic circuit under test is faulty. Online test means that the circuit to be tested is connected to the detection circuit. When all the input vectors are input from the input terminal through the entire detection circuit, it can directly determine whether the circuit under test is faulty by judging the output vector. Essentially, the proposed design in this paper is an online test model. Fault tolerance is to allow certain specific hardware or software errors in the circuit system while the system is still able to perform the specified functions.

For different fault models, including single stuck at fault of traditional digital circuits, commonly used methods are ILP, DFT, ATPG algorithm and so on. Most testing strategies could rely on the DFT. The testing circuit is designed for the specific circuit under test of logic functions. In a reversible circuit, for example, it is possible modified the reversible gates and enable them to generate information about the cascaded gates. By using this information, an appropriate reversible gate is added to detect failures like lost gate and duplicate gates [21]. After designing the DFT circuit, the test pattern sets should be defined for scan the circuit. ATPG algorithm is one common method to generate testing patterns for specific circuits. It generates as less pattern as possible to cover the maximum faults. Similarly, these methods can also be used for fault detection of quantum reversible circuits. Compared with ordinary digital circuits, quantum logic circuits have the advantages of controllability and observability. Controllability means that if a fault occurs in the circuit, changing one of the states in the circuit must also change the final output of the circuit. Observability is that a specific state of any layer in a logic circuit can be obtained through an input vector. Due to

the controllability and observability of the quantum logic circuit, it is possible to determine the fault condition of the quantum circuit by feeding a series of vectors to its input and observing the result of its output.

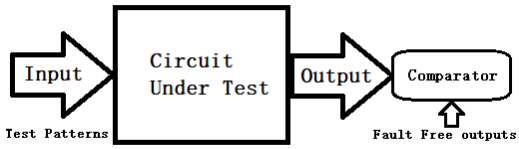


Figure 3. Black box testing model

IV. Proposed Solution and Mathematical Model

4.1 Testing model

The black box testing model of a digital circuit is described in Figure 3. The circuit is good or faulty with the given testing patterns by comparing the outputs to the saved fault-free outputs. This is done either online or offline. By using quantum reversible circuit, it allows to compare the input with the output directly. The proposed model is shown in Figure 4. Since the circuit under test is reversed by adding its reverse circuit. The output should produce exactly same pattern as the original inputs. The input-output mapping is traced in Figure 5.

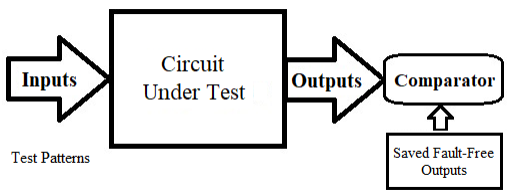


Figure 3. Black box testing model

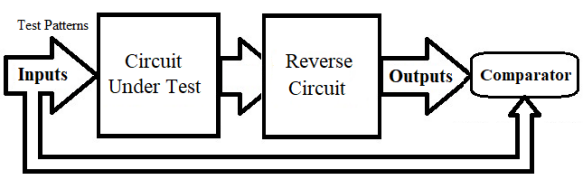


Figure 4. Black box testing model of a reversed circuit

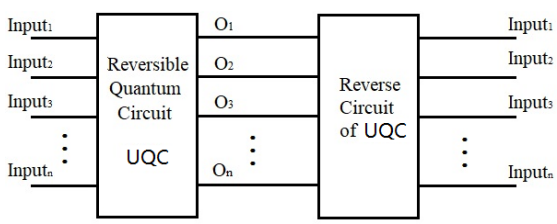


Figure 5. The input-output mapping of a reversed circuit

This model is described mathematically as follow (universal quantum logic circuit, UQC):

$$\boldsymbol{UQC}_{matrix} \times \boldsymbol{Input}_{matrix} = \boldsymbol{Output}_{matrix} \qquad (1)$$

$$\boldsymbol{UQC}_{matrix} \times \boldsymbol{UQC}^{-1} \times \boldsymbol{Input}_{matrix} = \boldsymbol{Input}_{matrix} \qquad (2)$$

There is an example of Toffoli gate that explains how reversible circuits detect faults in Figure 6 and 7:

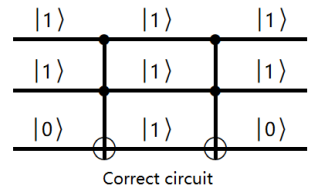


Figure 6. The Toffoli gate is self-reversed

In Figure 6, if input is |a, b, c> = |1, 1, 0>, the correct output after the gate is reversed by itself is still |1, 1, 0>. The corresponding matrix expression is:

$$\begin{bmatrix} 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0 \end{bmatrix} \times \begin{bmatrix} 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0 \end{bmatrix} \times \begin{bmatrix} 0\\0\\0\\0\\0\\0\\1\\0 \end{bmatrix} = \begin{bmatrix} 0\\0\\0\\0\\0\\0\\1\\0 \end{bmatrix}$$

Now, if there is a single stuck-at fault as shown in Figure 7, the line stuck at 0 will always produce |0> no matter what input is. It is detectable by adding its reversed logic gate (for the Toffoli gate, the reversed gate is itself) and comparing the output with the original input state.

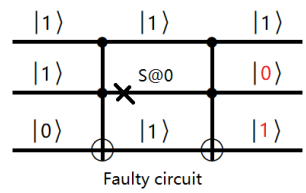


Figure 7. Detecting the stuck-at-fault of a Toffoli gate

The corresponding matrix expression:

$$\begin{bmatrix} 1&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&1&0&0&0&0&0\\ 0&0&0&1&0&0&0&0\\ 0&0&0&0&1&0&0&0\\ 0&0&0&0&0&1&0&0\\ 0&0&0&0&0&0&0&1\\ 0&0&0&0&0&0&1&0 \end{bmatrix} \times \begin{bmatrix} 1&0&1&0&0&0&0&0\\ 0&1&0&1&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&1&0&0&1\\ 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0 \end{bmatrix} \times \begin{bmatrix} 0\\0\\0\\0\\0\\0\\1\\0 \end{bmatrix} = \begin{bmatrix} 0\\0\\0\\0\\0\\1\\0\\0 \end{bmatrix}$$

The original input state is |1, 1, 0> and its matrix is $[0, 0, 0, 0, 0, 0, 1, 0]^T$. The reversed output state is |1, 0, 1> and its matrix is $[0, 0, 0, 0, 0, 1, 0, 0]^T$. The reversed output is different from the original input, thus there is a fault in the circuit. This is the fundamental principle that how the proposed fault self-detectable quantum circuit array works for fault detection.

4.2 Universal reversible quantum logic unit pair

The proposed universal reversible quantum logic unit pair is 4-qubit input and 4-qubit output quantum circuits and their structures are shown as Figure 8 and 9. The unit B is simply the mirror-flipped circuit of unit A. Certainly, the unit A is also the mirror-flipped circuit of unit B. Figure 10 shows that how unit A is reversed by adding unit B; and correspondingly, Figure 11 shows that unit B is reversed by adding unit A.

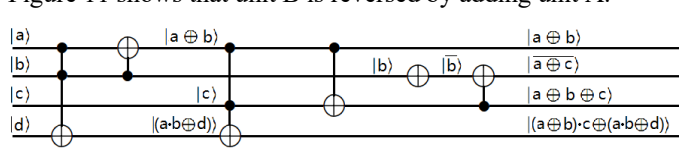


Figure 8. Universal reversible quantum logic: unit A

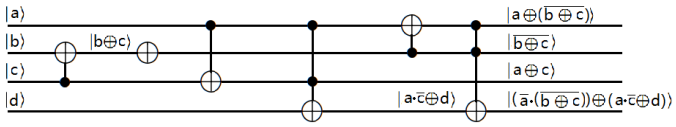


Figure 9. Universal reversible quantum logic: unit B

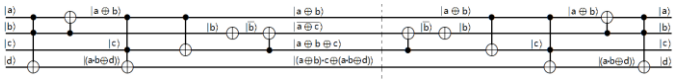


Figure 10. Unit A is reversed by unit B

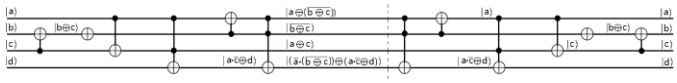
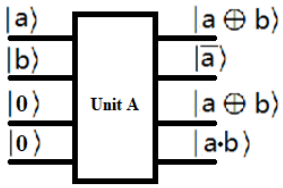
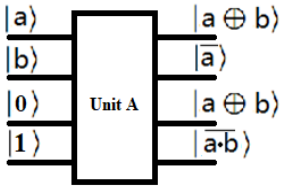


Figure 11. Unit A is reversed by unit B

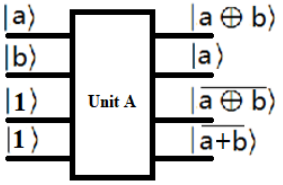
In this section, we define some important logic functions by proposed universal quantum unit A and B. Firstly, the function of unit A is given in Figure 12. By feeding different value to controlling bits |c> and |d>, we can have logic functions such as AND, NAND, OR, NOR, XOR, XNOR and Invertor. In addition, a full adder is simply set up by making the controlling bit |d> equal to be |0>. |a> and |b> are the input bits and |c> is the Carry-In. Then, the output at the bottom line becomes |ab+ac+bc> which is the logic for the Carry-Out. The output at third line becomes |a⊕b⊕c> which is the logic for Sum Accordingly, an *n*-bit full adder can be achieved by connecting *n*-unit A in a chain as shown in Figure 13.



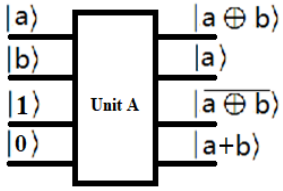
(a) AND and XOR gate



(b) NAND and Invertor



(c) NOR and XNOR gate



(d) OR and XNOR gate

Figure 12. Logic functions of Unit A by setting controlling bits |c> and |d> with 00, 01, 10, and 11.

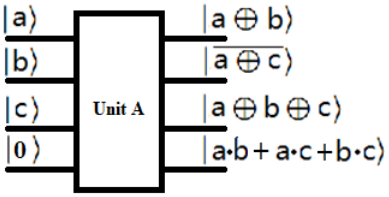


Figure 12. The 1-bit Full Adder by unit A

Similarly, the functions of unit B can be defined by feeding values to controlling bit such as logic XOR, XNOR and Invertor. Rather than these, unit B is also well defined as fan-outs shown in Figure 14. When |a> = |0>, |b> = |1> and |d> = |0>, unit B becomes a 4-branch fan-out for |c>. When |a> = |1>, |b> = |1> and |d> = |1>, unit B becomes a 4-branch fan-out for |c> and its invention. One unit B is possible to build a half-subtractor by setting |c> = |0> and |d> = |1>. In this case, |a> is the subtrahend; |b> is the minuend; the output at top line is the difference; the output at bottom line is the Sub_Out (Borrow). Further, the 1-bit full-subtractor can be achieved by connecting two unit B and one unit A in Figure 15.

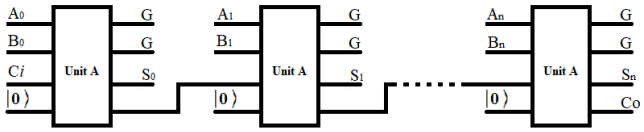
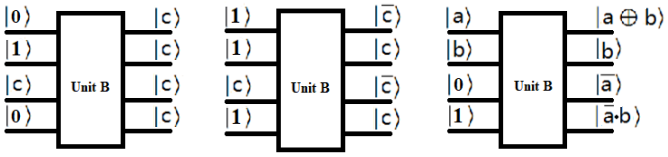


Figure 13. The *n*-bit full adder by unit A (G: garbage value)



(a) Fan-out: 1 (b) Fan-out: 2 (c) Half-Subtractor

Figure 14. Logic functions of Unit B

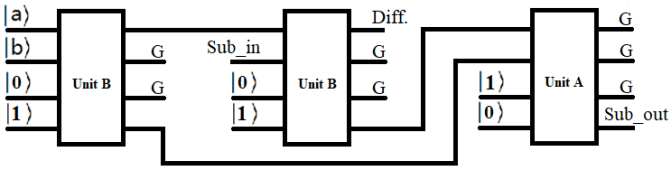
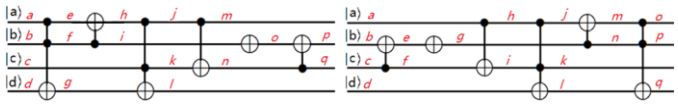


Figure 15. The 1-bit Full-Subtractor by connecting two unit B and one unit A (G: garbage value)

4.3 Single stuck-at-fault of the universal quantum units

Each universal quantum logic unit has 4 inputs. Therefore, the complete test set has $2^4 = 16$ patterns. While, there are 17 lines in each circuit unit. As a single stuck-at-fault occurs in each unit, it is one of two possible faults: stuck at logic 0 (S@0) or stuck at logic 1 (S@1). Thus, the total possible faults in each unit is 17×2 = 34. For each unit, each line is marked as it is shown in Figure 16. Finally, the single stuck-at-fault has been simulated by Python program in order to get the fault table.



(a) Unit A (b) Unit B

Figure 16. Mark lines for single stuck-at-fault simulation

V. Simulation Results and Model Establishment

The simulation of the single stuck-at-fault for each universal quantum logic unit was programmed in Python. With the complete test set of 16 patterns (0000, 0001 … 1111), we firstly plotted the truth table for each unit (table 1 and table 2). Then, we simulated the fault table to analyze the minimal test set.

Table 1. Truth table of universal quantum logic unit A

	abcd	Output	DEC.
0	0000	0100	4
1	0001	0101	5
2	0010	0010	2
3	0011	0011	3
4	0100	1110	14
5	0101	1111	15
6	0110	1001	9
7	0111	1000	8
8	1000	1010	10
9	1001	1011	11
10	1010	1101	13
11	1011	1100	12
12	1100	0001	1
13	1101	0000	0
14	1110	0111	7
15	1111	0110	6

Table 2. Truth table of universal quantum logic unit B

	abcd	Output	DEC.
0	0000	1101	13
1	0001	1100	12
2	0010	0010	2
3	0011	0011	3
4	0100	0000	0
5	0101	0001	1
6	0110	1111	15
7	0111	1110	14
8	1000	0111	7
9	1001	0110	6
10	1010	1000	8
11	1011	1001	9
12	1100	1011	11
13	1101	1010	10
14	1110	0100	4
15	1111	0101	5

The simulation included two cases. In first case, we simulated unit A as the faulty circuit and unit B as the testing circuit. In the second case, unit B was the faulty circuit and unit A was the testing circuit. Each simulation case needed total 544 loops by 34 faults × 16 patterns. As the simulation result, all faults are detectable by the complete test set. The minimal test pattern set for testing unit A (reversed by unit B) is (0000, 0111, 1110); while the minimal test pattern set for testing unit B (reversed by unit A) is (0010, 1101, 1111). In a simple word, it is possible to cover all single stuck-at-fault by using only three patterns in each case. However, the combination of the three the minimal test patterns are not unique in each case. The pattern has been selected if it covers as many faults as possible in simulation results.

5.1 The quantum reversible logic unit array

Now, we have designed the universal reversible quantum logic unit pair and simulated the single stuck-at-fault already. This section will discuss the fault detection in a large scale of the circuit built by the universal reversible quantum logic unit pairs. It is not need to add DFT for testing since the universal reversible quantum logic unit pair is designed for the self-testing[14]. The question right here is how to implement the testing without increasing the testing complexity.

Assumption:

As the assumption is made for testing simulation, in one complete fault detection, only one fault occurs. For a reliable CMOS VLSI circuit, the error rate is normally less than 1×10^{-12}.We assume that the quantum circuit is not as reliable as the CMOS circuit; however it is still a usable circuit whose error rate is about 1×10^{-3}. Therefore, it takes a very small chance that the multiple faults occur in the circuit at a time.

Assume that the defined quantum circuit system includes a number of *n* universal reversible quantum logic units in pair. One consideration is that we follow the common DFT method, which implies that the *n/2* of units are applied as the reversible circuit for testing. The problem is it is not an easy job to create the reversible circuit because of the complexity of connection circuit. We have to pay the extra cost for the complicated connections. Furthermore, the fault table is changed accordingly in this case and it depends on the inputs. If the defined quantum circuit system has *m* inputs, total possible testing patterns are 2^m. Consequently, the pattern set grows exponentially when the number of inputs increases. It would be better if we are able always to rely on the 3 same minimal test pattern for testing.

In the second consideration, we test the circuit by testing pair by pair. The method promises the 3 minimal test pattern for testing and the connection complexity is lower than that in first consideration. However, there are total *n* unit pairs in the quantum circuit system. It requires n/2 steps in the worst case to complete the testing. The testing complexity is *O(n)*; therefore, the testing complexity increases linear when the number of logic units increases. Obviously, this testing model benefits small circuit; however, it is not efficient for testing a large scale of the quantum logic circuit. It is necessary to reduce the testing complexity.

In order to reduce the testing complexity, we designed the quantum reversible logic unit array, represented in Figure 17. The unit A and B are arranged in pair alternately in rows and columns. Each row and column also have even number of the pairs. The connection circuit is switched between two modes: function mode and testing mode. The quantum reversible logic unit array performs its logic function in function mode. Correspondingly, the quantum reversible logic unit array

performs self-detection in testing mode. The idea on reducing the testing complexity is firstly to allocate n units into 2D array. Then, we connect all units as a chain in each row or in each column. Thus, if there is a fault in a row/column, the fault will be propagated to the end output through the chain of this row/column. Thereupon, it allows to scan row by row and column by column, instead of one pair by one pair. For n units in the array, each row has $\sqrt{n}$ unit and each column has $\sqrt{n}$ units. The total scan steps is $2\sqrt{n} = \sqrt{n} + \sqrt{n}$. Thus, the complexity in this case is $O(\sqrt{n})$ that is much better than $O(n)$.

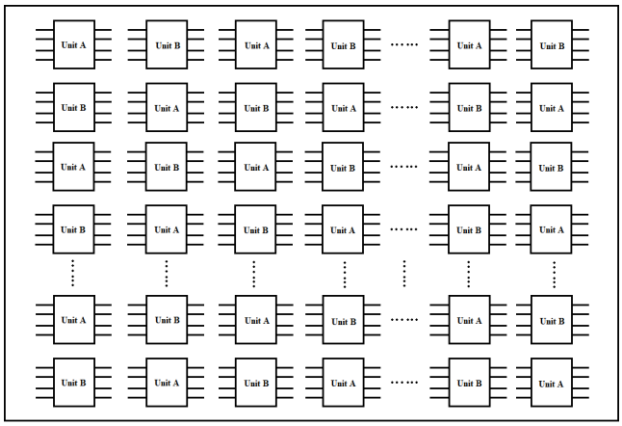


Figure 17. The proposed Quantum Reversible Logic Array

5.2 Self-Detection of the proposed logic unit array

The testing of the proposed logic unit array is efficient for a large scale quantum circuit because the testing complexity of the proposed logic unit array is $O(\sqrt{n})$. In this section, the implementation of the self-detection of the proposed logic unit array is explained as follow. The strategic steps are also shown in Figure 18.

Implementation:

Step 1. Switch to Row-Scan mode so that each unit is connected in a chain for each row.

Step 2. Implement testing for each row:

(1) if the input matches the output, move to next row;

keep scanning till the last row;

exit – no fault is detected.

(2) if the input does not match the output in the current row, mark the row as "detected fault" and move to the step 3.

Step 3. Switch to Column-Scan mode so that each unit is connected in a chain for each column.

(1) if the input matches the output, move to next column; keep scanning till finding that the input does not match the output in a column; then, mark it as "located fault". The faulty unit is located at the cross of the row "detected fault" and the column " located fault";

(2) exit – the fault is detected.

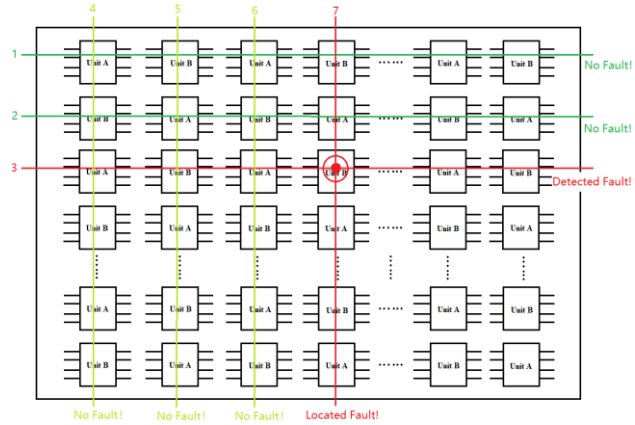


Figure 18. Fault detection of the proposed Quantum Reversible Logic Array

5.3 Comparison

The existing testing method in a quantum logic circuit is usually based on the ATPG [22-25]. Firstly, the size of input pattern sets for testing depend on the number of inputs, and test patterns need to be updated with different circuit constructs. It does not matter the testing is focus on a single gate or a universal logic unit. In addition, even their design is optimized for enhancing testing efficiency, when the inputs in the circuit increase, the complexity of input pattern sets increase linearly anyway. Secondly, input pattern sets generated by ATPG algorithm in some certain circuits cannot guaranty a full fault coverage, which means some faults are not detectable in some case. Thirdly, the related works also involved extra DFT circuit for testing the circuit [26-28]. DFT increases hardware overhead and it normally introduces garbage value during the testing process; thus, the test cost of the entire circuit increases. Parity preserving [29, 30] has similar problem. A parity preserving circuit is always verified using a specific fault model and does not ensure reasonable the fault coverage automatically. So far, its benefits to quantum reversible circuits have been limited.

The proposed design in this paper has two universal logic units that can build more complex functional circuits. It has the fault self-detectable mechanism and does not require any DFT circuit for testing purpose. Therefore, the testing cost becomes much lower and there is no garbage value produced. The testing pattern sets are fixed and will not change with different circuit constructs. More importantly, the testing complexity of our design is $O(\sqrt{n})$ instead of increasing linearly, that is much more efficient than the other mentioned works in the large scale circuit. This feature greatly benefits the faults testing procedure of large quantum circuits compared to other existing methods.

VI. CONCLUTION

Our proposed fault self-detectable universal quantum logic unit array is based on two facts; the universal quantum logic unit is the block to build a complex logic functional circuit and additionally, the quantum circuit is reversible, so detecting a fault can be achieved by adding its reversed circuit and comparing the output with the original input. In this paper, we

have introduced and designed a novel universal quantum unit array that allows fault self-detection by using 3 test patterns as a minimal test pattern set. The proposed low power design of reversible circuit does not use individual DFT model for the fault detection. It is not a complex connection circuit for testing. The testing complexity is $O(\sqrt{n})$ that is efficient for a large scale quantum circuit testing. At last, the testing model in this paper is limited on detecting a single faulty unit. It is possible to extend the model to detect the multiple faulty units without changing the complexity. However, some faulty units will result in an undetectable fault by an extremely small chance in testing. Moreover, this will require the modification and more additional complex connection circuit. Since the probability of the multiple faults is very low in reality, the paper does not include the discussion on that case.

As our future work, the optimized two-dimensional fault self-detectable universal quantum logic unit array will be exploited for even more efficient self-detection. Also, based on the low power consumption characteristics of quantum circuits, it is possible to construct a three-dimensional circuit. In this case, the proposed fault self-detectable universal quantum logic unit array can be extended into a three-dimensional unit block. Eventually, the self-detection algorithm can be also developed for testing the three-dimensional unit block. The design with the corresponding testing method will be deeply studied in future.

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Appendix A

Figure a. Screenshots of partial simulation result.

Table a-1. Simulation result for each S@F in unit A (part 1)

A	abcd	a: s@1	e:s@1	h:s@1	j:s@1	m:s@1	a: s@0	e:s@0	h:s@0	j:s@0	m:s@0	b:s@1	f:s@1	i:s@1	o:s@0	p:s@0	b:s@0	f:s@0	i:s@0	o:s@1	p:s@1
0	0000	1000	1000	1000	1000	1011	0000	0000	0000	0000	0000	0100	0100	1101	1101	1101	0000	0000	0000	0000	0000
1	0001	1001	1001	1001	1001	1010	0001	0001	0001	0001	0001	0101	0101	1100	1100	1100	0001	0001	0001	0001	0001
2	0010	1010	1010	1010	1011	1000	0010	0010	0010	0010	0010	0110	0110	1111	1111	0010	0010	0010	0010	0010	1111
3	0011	1011	1011	1011	1010	1001	0011	0011	0011	0011	0011	0111	0111	1110	1110	0011	0011	0011	0011	0011	1110
4	0100	1100	1101	0100	0100	0100	0100	0100	1101	1101	1111	0100	0100	0100	0100	1000	0000	0000	1000	1000	0100
5	0101	1101	1100	0101	0101	0101	0101	0101	1100	1100	1110	0101	0101	0101	0101	1001	0001	0001	1001	1001	0101
6	0110	1110	1111	0110	0110	0110	0110	0110	1111	1110	1100	0110	0110	0110	0110	0110	0010	0010	1010	1010	1010
7	0111	1111	1110	0111	0111	0111	0111	0111	1110	1111	1101	0111	0111	0111	0111	0111	0011	0011	1011	1011	1011
8	1000	1000	1000	1000	1000	1000	0000	0000	0000	0000	0010	1100	1101	0100	0100	1000	1000	1000	1000	1000	0100
9	1001	1001	1001	1001	1001	1001	0001	0001	0001	0001	0011	1101	1100	0101	0101	1001	1001	1001	1001	1001	0101
10	1010	1010	1010	1010	1010	1010	0010	0010	0010	0011	0001	1110	1111	0110	0110	0110	1010	1010	1010	1010	1010
11	1011	1011	1011	1011	1011	1011	0011	0011	0011	0010	0000	1111	1110	0111	0111	0111	1011	1011	1011	1011	1011
12	1100	1100	1100	0101	0101	0110	0100	0101	1100	1100	1100	1100	1100	1100	1100	1100	1000	1001	0001	0001	0001
13	1101	1101	1101	0100	0100	0111	0101	0100	1101	1101	1101	1101	1101	1101	1101	1101	1001	1000	0000	0000	0000
14	1110	1110	1110	0111	0110	0101	0110	0111	1110	1110	1110	1110	1110	1110	1110	0011	1010	1011	0011	0011	1110
15	1111	1111	1111	0110	0111	0100	0111	0110	1111	1111	1111	1111	1111	1111	1111	0010	1011	1010	0010	0010	1111

Table a-2. Simulation result for each S@F in unit A (part 2)

A	abcd	c:s@1	k:s@1	n:s@1	q:s@1	c:s@0	k:s@0	n:s@0	q:s@0	d:s@1	g:s@1	l:s@1	d:s@0	g:s@0	l:s@0
0	0000	0010	0010	0010	1111	0000	0000	0000	0000	0001	0001	0001	0000	0000	0000
1	0001	0011	0011	0011	1110	0001	0001	0001	0001	0001	0001	0001	0000	0000	0000
2	0010	0010	0010	0010	0010	0000	0000	0000	1101	0011	0011	0011	0010	0010	0010
3	0011	0011	0011	0011	0011	0001	0001	0001	1100	0011	0011	0011	0010	0010	0010
4	0100	0110	0111	0100	0100	0100	0100	0111	1011	0101	0101	0101	0100	0100	0100
5	0101	0111	0110	0101	0101	0101	0101	0110	1010	0101	0101	0101	0100	0100	0100
6	0110	0110	0110	0101	1001	0100	0101	0110	0110	0111	0111	0110	0110	0110	0111
7	0111	0111	0111	0100	1000	0101	0100	0111	0111	0111	0111	0110	0110	0110	0111
8	1000	1010	1011	1000	1000	1000	1000	1011	0111	1001	1001	1001	1000	1000	1000
9	1001	1011	1010	1001	1001	1001	1001	1010	0110	1001	1001	1001	1000	1000	1000
10	1010	1010	1010	1001	0101	1000	1001	1010	1010	1011	1011	1010	1010	1010	1011
11	1011	1011	1011	1000	0100	1001	1000	1011	1011	1011	1011	1010	1010	1010	1011
12	1100	1110	1110	1110	0011	1100	1100	1100	1100	1101	1100	1100	1100	1101	1101
13	1101	1111	1111	1111	0010	1101	1101	1101	1101	1101	1100	1100	1100	1101	1101
14	1110	1110	1110	1110	1110	1100	1100	1100	0001	1111	1110	1110	1110	1111	1111
15	1111	1111	1111	1111	1111	1101	1101	1101	0000	1111	1110	1110	1110	1111	1111

The green area: Top line, the yellow area: Second line, the gray green area: Third line and the light purple area: Bottom line.